

Experiment No. 3

Experiment Name

Statistical Analysis and Hypothesis Testing using Python

Aim

To perform statistical analysis and hypothesis testing on a real-world dataset using Python in order to identify relationships between variables, validate assumptions, and support data-driven decision-making.

Purpose

Statistical analysis enables data analysts to summarize data, identify trends, measure relationships, and test hypotheses using mathematical techniques. It plays a crucial role in determining whether observed patterns are statistically significant or occurred by chance.

In this experiment, students will work with datasets such as the IBM HR Employee Attrition Dataset, Titanic Survival Dataset, or House Price Prediction Dataset to perform descriptive statistics, correlation analysis, hypothesis testing (t-test and ANOVA), and regression analysis. The results will help students draw meaningful conclusions and make evidence-based decisions.

Prerequisites

- Basic knowledge of Python programming
- Familiarity with Pandas, NumPy, SciPy, and Statsmodels libraries
- Understanding of descriptive statistics (mean, median, variance, standard deviation)
- Basic concepts of probability and hypothesis testing
- Jupyter Notebook or Google Colab

Steps

1. Select a real-world dataset such as the IBM HR Employee Attrition Dataset, Titanic Survival Dataset, or House Price Prediction Dataset.
2. Import the required Python libraries including Pandas, NumPy, SciPy, Matplotlib, and Statsmodels.
3. Load the dataset and calculate descriptive statistics such as mean, median, mode, variance, and standard deviation for relevant numerical features.
4. Analyze relationships between variables by calculating the Pearson correlation coefficient and visualizing the correlation matrix.

5. Formulate a Null Hypothesis (H_0) and an Alternative Hypothesis (H_1) for a business problem (e.g., "Is there a significant difference in monthly income between employees who leave the company and those who stay?").
6. Perform an Independent Sample t-test to compare the means of two groups and interpret the p-value.
7. Apply One-Way ANOVA to determine whether there is a statistically significant difference among multiple groups (e.g., employee satisfaction across different job roles).
8. Build a simple Linear Regression model to analyze the relationship between two numerical variables, such as Years of Experience and Monthly Income or House Area and House Price.
9. Interpret the statistical outputs, including p-values, confidence intervals, regression coefficients, and R^2 score.
10. Summarize the findings and discuss how statistical analysis can support business decisions and predictive analytics.

Questions

1. What is the difference between descriptive statistics and inferential statistics?
2. Explain the concepts of the Null Hypothesis (H_0) and Alternative Hypothesis (H_1).
3. What is a p-value? How is it used to make statistical decisions?
4. Differentiate between a t-test and ANOVA. In which situations is each test applied?
5. What does the Pearson correlation coefficient indicate? What are its possible values?
6. Explain the significance of R^2 (Coefficient of Determination) in Linear Regression.
7. Why is statistical analysis important before applying machine learning algorithms?
8. What assumptions should be satisfied before performing a t-test or ANOVA?
9. How can regression analysis help organizations in forecasting and decision-making?
10. Give two real-world applications where hypothesis testing is commonly used in data analytics.